MILITARY SPECIFICATION

CODE CARD, TRAINING

1. SCOPE

1.1 This specification covers special requirements not included on the drawings, and provides the quality assurance provisions for the fabrication of parts, assembling, and the preparation for delivery for one type of code card used with the Atomic Demolition Charge, Training and designated as Code Card, Training.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-T-90 - Tape, Pressure Sensitive, Adhesive (Cellophane and Cellulose Acetate)

MILITARY

MIL-B-22205 - Bags, Transparent, Flexible, Heat Sealable for Packing Applications
MIL-P-116 - Preservation, Methods of
MIL-P-117 - Bag and Sleeve, Interior Packaging
MIL-P-13949 - Plastic Sheet, Laminated, Copper Clad for Printed Wiring

STANDARDS

FEDERAL

Federal Test Method Standard No. 151
Metals: Test Method

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition

DRAWINGS

ORDNANCE CORPS

8808362 - Packing and Marking for Storage and Shipments
8851882 - Code Card, Wire Test Set A, Training
8851883 - Code Card, Wire Test Set B, Training
8851884 - Code Card, Radio Test Set A, Training
8851885 - Code Card, Radio Test Set B, Training
8851897 - Code Card, Radio Training
8851898 - Code Card, Wire Training
8851921 - Box, Set-Up Paperboard, for Code Cards Training
8851945 - Box, Set-Up Paperboard, for Code Cards
8852053 - Code Cards Radio
8852054 - Code Cards Wire
8854416 - Electrical Test Panel
8854423 - Operating Instructions for 8854416
8886568 - Box, Packing, Ammunition, for Code Card Set
8886569 - Box, Packing, Ammunition, for Code Card Set
EL-8851882 - Equipment List
EL-8851883 - Equipment List
EL-8851884 - Equipment List
EL-8851885 - Equipment List
EL-8851897 - Equipment List
EL-8851898 - Equipment List

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material used to fabricate any part complies with the specifications listed on the applicable drawing.

3.2 Continuity. The code card shall have a continuity in accordance with the applicable drawing when tested as specified in 4.4.1.

3.3 Insulation resistance. The code card shall have an insulation resistance of 30 megohms, min. when a potential of 500 ± 50 volts direct current (Vdc) with current limited to 5 milliamperes (ma) is applied as specified in 4.4.2.

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3.4 Paint and conductor adhesion. The plated and conductive surfaces of the code card shall show no evidence of peeling, chipping, or flaking when tested as specified in 4.4.3.

3.5 Peel strength. The code card shall have a minimum peel strength of 9 pounds per inch width as determined by testing in accordance with 4.4.4.

3.6 Serialization. Serial numbers shall be assigned by the procuring agency. There shall be duplication of serial numbers assigned each type of card (see applicable drawing).

3.7 Workmanship.

3.7.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.7.2 Burrs. Parts shall be free from burrs.

3.7.3 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.7.4 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination.

A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected

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for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Major characteristic table I

Lot size	Sample size
1- 25	100% inspection
26- 30	25
31- 35	26
36- 40	27
41- 45	28
46- 50	29
51- 55	30
56- 60	31
61- 65	32
66- 70	33
71- 75	34
76- 85	35
86- 95	36
96- 110	37
111- 130	38
131- 156	39
157- 190	40
191- 240	41
241- 340	42
341- 520	43
521-1120	44
1121 and over	45

Minor characteristic table II

Lot size	Sample size
1- 25	100% inspection
26- 35	16
36- 45	17
46- 55	18
56- 65	19
66-100	20
101-200	21
201 and over	22

4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Code card, wire test set A training (see dwg 8851882), code card, wire test set B, training (see dwg 8851883), code card, radio test set A (see dwg 8851884), code card radio test set B, training (see dwg 8851885), code card radio training (see dwg 8851897), and code card wire training (see dwg 8851898).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:

* 101. Location of registry (both sides).....Gage

* 102. Outline of registry (both sides).....Gage

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103.	Peeling or separation of conductor from board	Visual
104.	Any looseness of bond including bubbles or blisters	Visual/Manual
105.	Bottom length to end notches	Gage
106.	Perpendicularity of bottom end slots	Gage
107.	Length from left bottom notch to left side of slot right of center slot	Gage
108.	Width of left bottom notch	Gage
109.	Length from left bottom notch to left side of slot left of center slot	Gage
110.	Width of 2 large bottom slots	Gage
111.	Location of right registry hole if present	Gage
112.	Diameter of right registry hole if present	Gage
113.	Depth of 2 end notches	Gage
114.	Total width	Gage
115.	Location of left registry hole if present	Gage
116.	Diameter of left registry hole if present	Gage
117.	Length from bottom end notches to sides of center slot	Gage
118.	Total length	Gage
119.	Thickness	Gage
120.	Depth of slots	Gage
121.	Circuitry clearly defined	Visual
122.	No warpage	Visual
123.	Smooth edges	Manual
124.	No copper flashing in slots and on edges	Visual
125.	Board not cracked or chipped	Visual
126.	Etched marking proper and identifiable	Visual
127.	No cuts tears or cracks in conductor portion	Visual
128.	Painting	Visual
129.	Paint proper color	Visual
130.	No paint on undesignated area	Visual
131.	Serialization (see 3.6)	Visual
132.	Materials (see 3.1)	Visual
133.	Workmanship (see 3.7)	Visual/Manual

* Each 35 or less code cards produced shall be inspected as follows: One card shall be inspected completely and the remaining 34 cards or less shall be inspected for the location and width of one contact on each portion of both sides (a total of 4 contacts) of the cards.

Minor: None defined.

4.3.2.1.2 Code cards radio (see dwg 8852053) and code cards wire (see dwg 8852053).

Categories	Characteristics	Method of inspection

Critical: None defined.

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Major:
101. Number of code cards.....Visual/Manual

Minor: None defined.

4.3.2.1.3 Box, set-up, paperboard for code cards (see dwgs 8851921, 8851945, 8886568, 8886569 and section 5).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Packaging materials (see 3.1).....Visual
102. Proper packing of 2 cards in inner bag.....Visual/Manual
103. Serial numbers.....Visual
104. Inner bag sealed in accordance with 3.2 MIL-B-2205.....Visual/Manual
105. Contents tight in box.....Manual
106. Box sealed with sealing strip.....Visual
107. Lable on box.....Visual
108. Marking on box label.....Visual
109. Corners blunted on box.....Visual
110. Air exhausted from bag.....Manual
111. Objective evidence (see 6.2) available for heat seal test of bag in accordance with MIL-P-116.....Visual
112. No hole, crack or split in bag.....Visual

Minor:
201. Marking of bag.....Visual
202. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.4 Packing and Marking for Storage and Shipments (see dwg 8808362 and section 5).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Gross load weight, max.....*Measure
102. Contents tight in package.....Manual
103. Correctly packed.....Visual/Manual
104. Package material (see 3.1).....Visual

*Determine weight with a suitable scale.

Minor:
201. Marking.....Visual
202. Workmanship (see 3.7).....Visual/Manual

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4.3.3 Testing. The contractor shall perform all the tests specified in this paragraph. Tests do not have to be performed in the order specified.

	Test	Test classification
a.	Continuity	Major
b.	Insulation resistance	Major
c.	Paint and conductor adhesion	Major
d.	Peel strength	Major

Test d is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Continuity (see 3.2) and insulation resistance (see 3.3). Each code card shall be subjected to these tests (see 4.4.1 and 4.4.2 as applicable) in the order specified. If any card fails to comply with 3.2 and 3.3, it shall be rejected and removed from the lot.

4.3.3.2 Paint and conductor adhesion (see 3.4). Each code card shall be subjected to test (see 4.4.3). If any card fails to comply with 3.4, it shall be rejected and removed from the lot.

4.3.3.3 Peel strength (see 3.5). One test specimen (see fig 1) shall be processed each day the code cards are produced and tested in accordance with 4.4.4. If any specimen fails to comply with MIL-P-13949, the lot represented by the specimen shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists(EL 8851882, 8851883, 8851884, 8851885, 8851897, and EL 8851898 as applicable). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Continuity. Continuity shall be determined in accordance with Continuity Test Fixture dwg 8854416, applicable operating instructions dwg 8854423, and the coding instructions furnished for each particular serialized set of card assemblies (see 6.4).

4.4.2 Insulation resistance. Insulation resistance shall be determined in accordance with Table III, Insulation Resistance Test Fixture dwg 8854416, applicable operating instructions dwg 8854423, and the coding instructions furnished for each particular serialized set of card assemblies (see 6.4).

Table III

Insulation resistance

Positive potential	Negative potential to all other points listed in "From" column
Safe buss on forward side	
Safe buss on rear side	
Fire buss on forward side	
Fire buss on rear side	
Each missing or broken bit	

4.4.3 Paint and conductor adhesion. The adhesion shall be determined by utilizing a strip of pressure-sensitive cellophane tape conforming to type I, Class A, of Specification L-T-90, 1/2 inch and 5 inches long (effective length). The tape shall be pressed firmly across the surface of the conductor pattern or the painted surface and then be removed rapidly by manual force applied approximately perpendicular to the pattern. Each side of the board shall be tested, one side shall be utilized for paint adhesion and the other side for plating adhesion. Fresh tape shall be used for each test.

4.4.4 Peel strength. The peel strength of the conductor shall be obtained in accordance with the test specified in Specification MIL-P-13949. This test shall be performed on the 4 applicable portions of the test specimen (see fig. 1).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The code cards shall be packaged in accordance with dwgs 8851921, 8851945, 8886568, 8886569, and 8808362 as applicable using the specified moisture barrier.

5.1.2 Level C. The code cards shall be packaged in accordance with dwgs 8851921, 8851945, 8886568, 8886569 and 8808362 as applicable except that the moisture barrier may be omitted.

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5.2 Packing.

5.2.1 Level A and C. The code cards shall be packed in accordance with dwgs 8851921, 8851945, 8886568, 8886569 and 8808362 as applicable

5.3 Marking. Marking shall be in accordance with dwgs 8851921, 8851945, 8886568, 8886569 and 8808362 as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

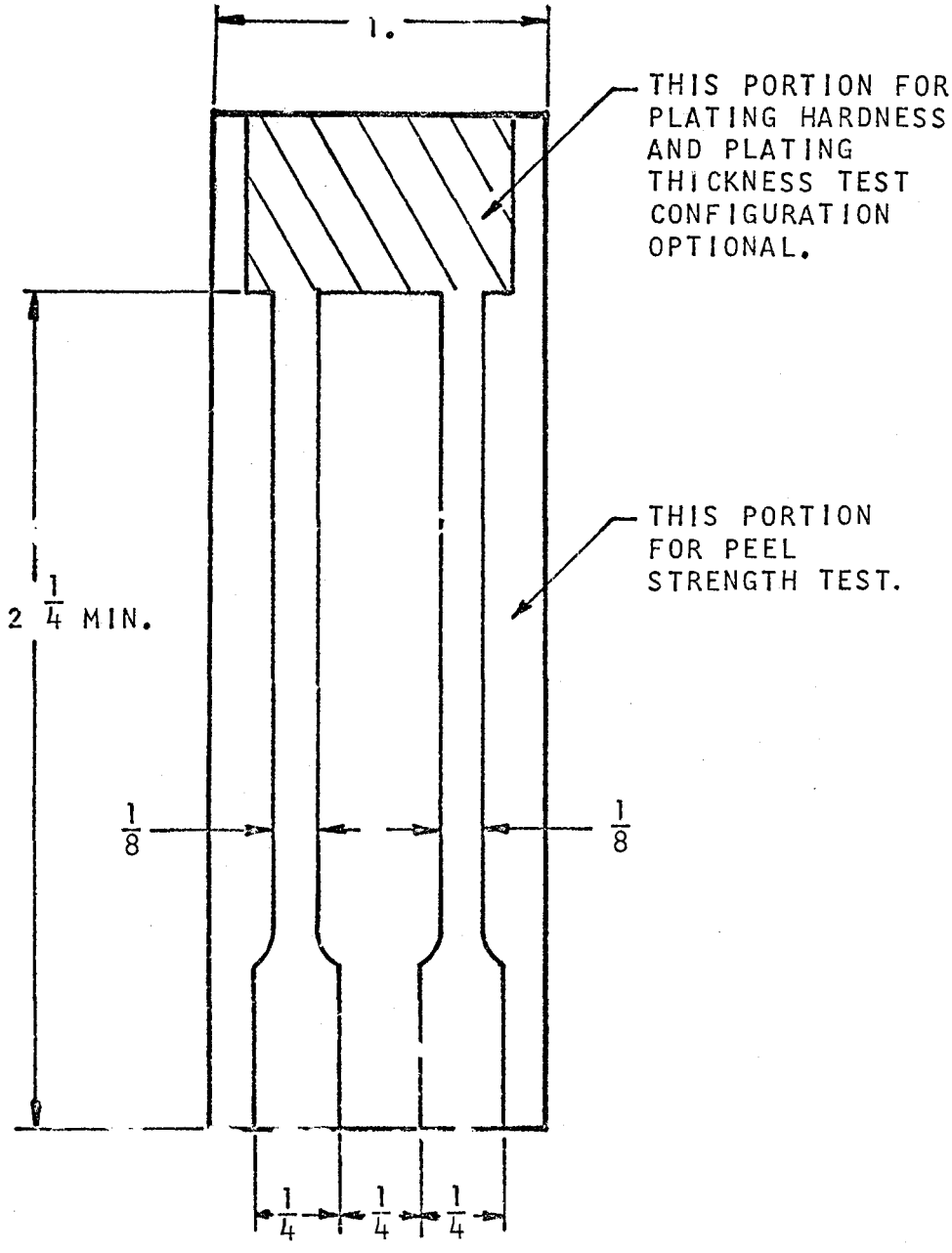
6.4 Assignment of codes and serial numbers. The procuring activity assigns codes and serial numbers.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1195-A049

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MATERIAL:

PLASTIC SHEET, LAMINATED, COPPER-CLAD, TYPE FL-G-C-2/2, SPEC MIL-P-13949.

NOTE: DUPLICATE PATTERN ON OPPOSITE SIDE OF BOARD.

FIGURE 1

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DEPARTMENT OF THE ARMY

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-50223(MU) Code Card, Training

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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